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EXPERIMENT 4
Finite-State Machine for Morphological Parsing

Aim

To implement a simple finite-state machine in Python that generates the plural form of an English noun.

Algorithm

• Read a singular noun from the user.

• Check whether the noun ends with s, x, z, ch, or sh. If so, add “es”.

• Otherwise, check whether the noun ends with “y” and the preceding letter is a consonant. If so, replace “y” with “ies”.

• For all other nouns, add “s”.

• Display the plural form.

Code

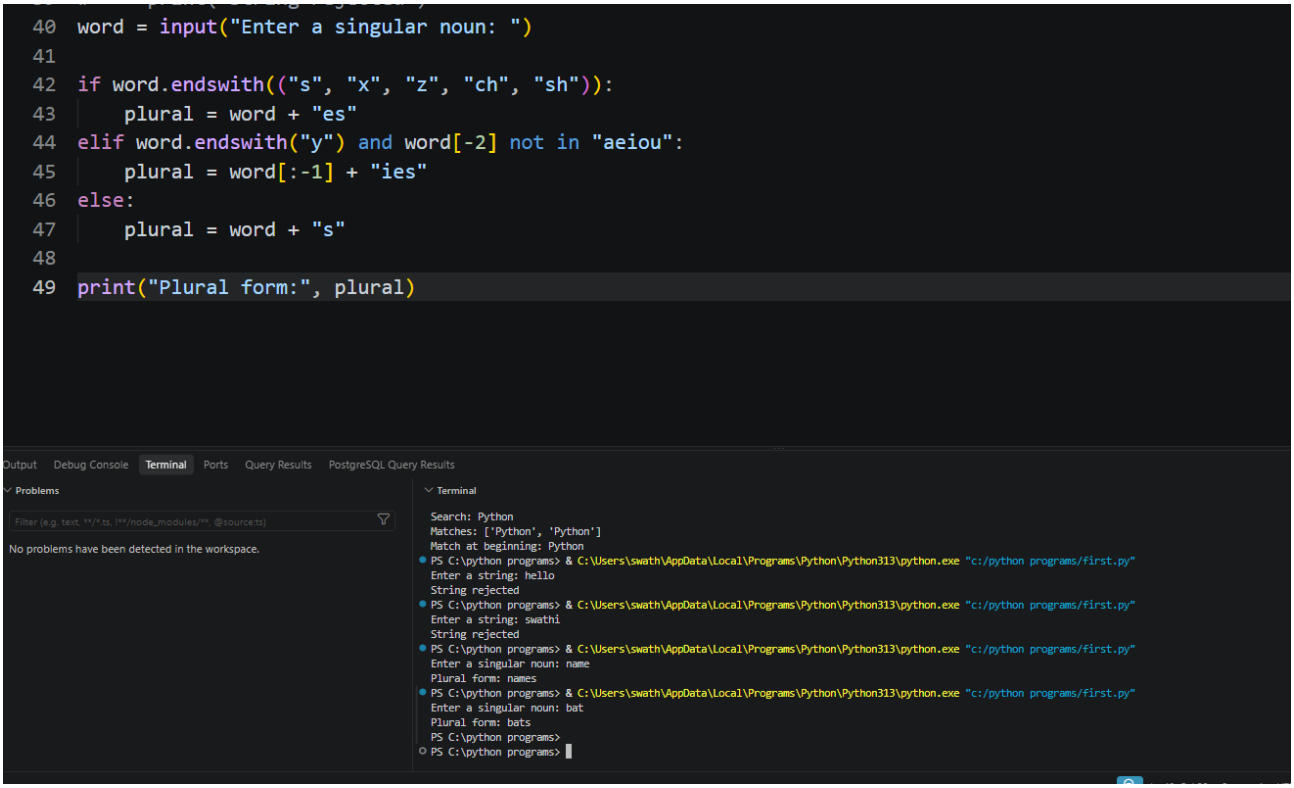
```
word = input("Enter a singular noun: ")

if word.endswith(("s", "x", "z", "ch", "sh")):
    plural = word + "es"
elif word.endswith("y") and word[-2] not in "aeiou":
    plural = word[:-1] + "ies"
else:
    plural = word + "s"

print("Plural form:", plural)
```

Output

Output screenshot provided:



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Result

The program generates plural forms for English nouns using simple suffix rules. For example, “name” becomes “names” and “bat” becomes “bats”.